

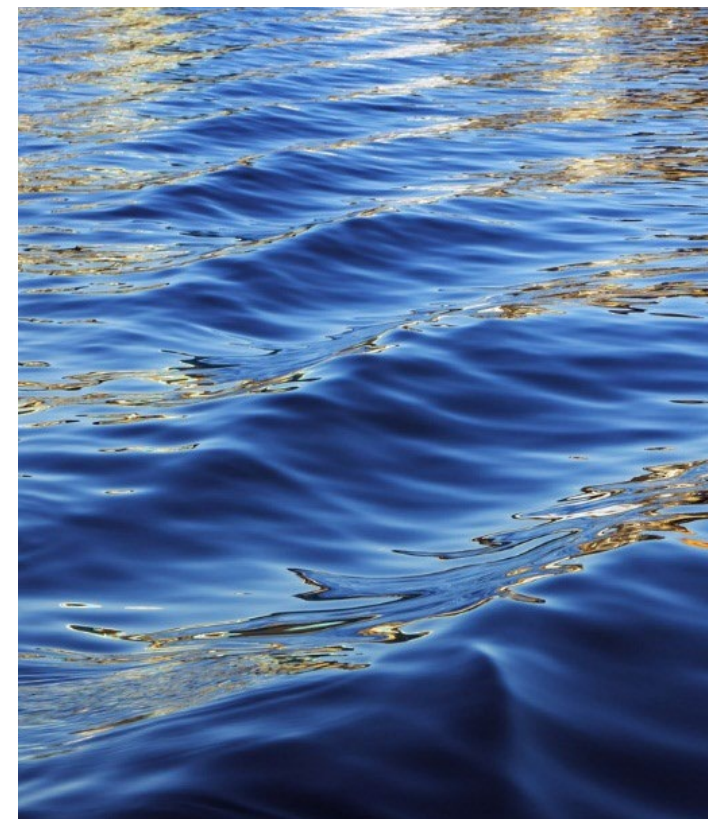


Risk Management

Operational risk management



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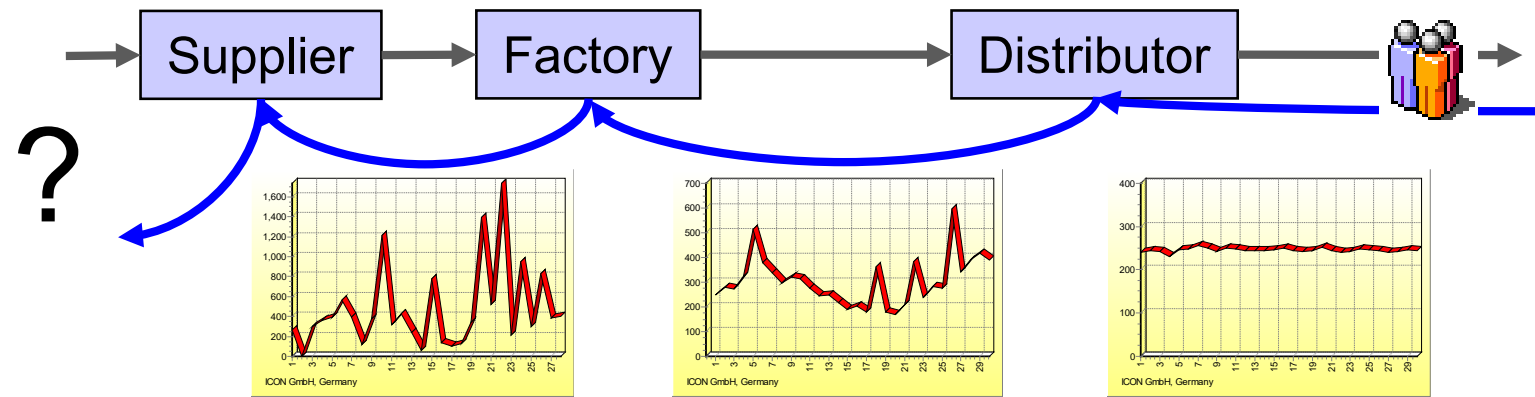


Learning objectives

- Describe the Bullwhip effect, causes and effects.
- How to detect and analyze.
- Illustrative examples for bullwhip effect
- Mitigation strategies: vendor-managed inventory.



The bullwhip effect



Bullwhip-effect

← Information flow
→ Material flow

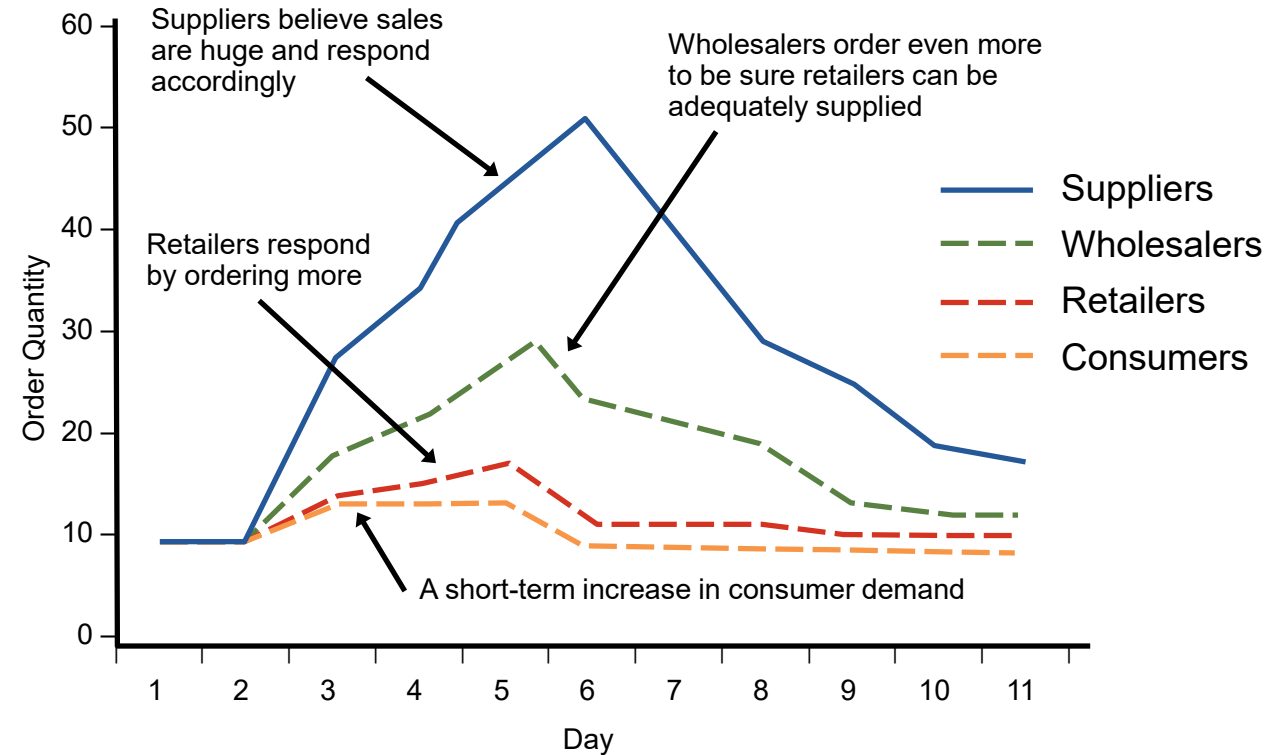
The magnification of order variability in the supply chain:

→ little variability in retail orders ... can lead to greater variability for fewer wholesalers, and ... can lead to even greater variability for a single manufacturer.



The Bullwhip effect

- The tendency for greater fluctuations in order size as orders are passed through the supply chain



- Creates unstable production schedules, costly capacity changes, longer lead times, obsolescence, and costly downtime.
- Damage can be minimized with supplier coordination and planning.



The bullwhip effect: causes

Operational causes

- Demand forecasting: inventory adjustment based on forecasts in various elements of the chain.
- Delivery time variability
- Batch/order synchronization
- Orders in large batches
- Discounts and promotions incentivizing quantity increases
- Trade promotion and forward purchase
- Anticipation of shortages.

Behavioral causes

- Misuse of stock policies
- Misperceptions of feedback and time delays
- Panic order reactions after unsatisfied demand
- Risk perceived by the limited rationality of other players.



The bullwhip effect: consequences

- **Excess inventory:**

Manufacturers or distributors may accumulate too much stock, which increases storage costs and the risk of obsolescence.

- **Stockout, shortage due to lack of products**

Companies can run out of inventory at critical times due to misinterpretation of demand, resulting in lost sales and customers.

- **Additional costs:**

Fluctuation in demand can lead to adjustments in production, generating additional manufacturing or supply costs (such as overtime or subcontracting).

- **Inefficiency in the chain:**

The entire supply chain becomes less efficient, with **longer lead times**, **coordination difficulties**, and **increased risk of errors**.



Example of bullwhip effect

“PELUCHITOS:”

- Manufacturer of dolls for sale in retail stores, through a distributor.
- The distributor adjusts its stock based on retailer order forecasts.
- Retail outlets adjust their orders to the company based on their forecasts and promotions.

SCENARIO 1:

- Suppose that end-consumer demand varies slightly in toy stores, declining a little (10%).

REACTIONS:

1. **In-store demand:**
 - Consumers buy an average of 100 dolls per week, but during a particular month, demand drops slightly to 90 dolls.
2. **Retailer response:**
 - Seeing this drop in demand, he anticipates that the trend will continue and reduces his order to the toy company from 120 to 80 dolls per week (based on his forecast).
3. **Distributor's response:**
 - Noticing a reduction in orders and, anticipating less demand, he also reduces his order to the toy company to only 60 dolls per week.
4. **Response from the toy company:**
 - The toy company, noticing this reduction in orders, decides to reduce production and order fewer materials from its suppliers to avoid accumulating inventory. It goes from manufacturing 150 to only 40 dolls per week, which affects its plastic and textile suppliers.





Example of bullwhip effect

REACTIONS:

1. **In-store demand:** Consumers buy an average of 100 dolls per week, but during a particular month, demand drops slightly to 90 dolls.
2. **Retailer's response:** The retailer, seeing this drop in demand, anticipates that the trend will continue and reduces its order to the toy company from 120 to 80 dolls per week (based on its forecast).
3. **Distributor's response:** The distributor supplying the retailer notices a reduction in orders and, anticipating less demand, also reduces its order to the toy company to only 60 dolls per week.
4. **Toy company's response:** The toy company, noticing this reduction in orders, decides to reduce production and order fewer materials from its suppliers to avoid accumulating inventory. It goes from manufacturing 150 to only 40 dolls per week, which affects its plastic and textile suppliers.

END EFFECT:

- The final consumer only reduced his purchase from 100 to 90 dolls per week (a variation of 10%).
- Due to the bullwhip effect, the toy company reduced its production by 60%, affecting upstream suppliers.
- If demand rises again, the toy company may not have enough inventory to meet needs, resulting in losses.



Example 2 of bullwhip effect



SCENARIO 2:

1. In-store demand:

- Consumers continue to buy an average of 100 dolls per week in stores.
- Because of a paid social media campaign or a popular event, retailers anticipate that there will be an increase in demand for dolls.

REACTIONS:

1. Retailer response:

- Anticipating this supposed increase in demand, they decide to increase their orders from 120 dolls to 150 dolls per week.

2. Distributor's response:

- Noticing the increase in retailer orders, it also adjusts its inventory.
- They decide to place an even larger order, in case demand continues to rise, and increase their orders to the toy company from 150 dolls to 200 dolls per week.

3. Response from the toy company:

- The toy company, receiving these larger orders from the distributor, interprets that demand is increasing dramatically.
- They decide to increase their production from 150 dolls to 300 dolls per week to meet demand and avoid running out of stock in the future.
- In addition, the company orders more materials from its plastic and textile suppliers, generating more raw material costs.

4. Suppliers:

- Noticing the increase in orders, they also adjust their production to meet the growing demand of the toy company, increasing their own production capacities.





Manage the bullwhip effect

Cause of the bullwhip effect	Countermeasures
Non-transparent claim	Coordination of information
Disregarding orders in an attempt to reduce inventory	Automated ordering and inventory monitoring to avoid overstocking or shortages
Batch order processing	Coordinated and precise definition of lot size
Promotions	Use of everyday low prices (EDLP) in lieu of promotions
Shortage gaming	Validation of customer demand through historical customer order data
Product returns	Policies to control returns or cancelled orders.



Analysis of the bullwhip effect

$$OrdVarR = \frac{\sigma_{order}^2 / \mu_{order}}{\sigma_{demand}^2 / \mu_{demand}}$$

where variations in demand and orders are denoted by σ , and μ are the average demand and orders.

If the OrdVarR measure is:

- > 1 - Variance amplification is present
- = 1 - No amplification is present
- < 1 - Damping is occurring



Bullwhip effect: example 2

- Imagine we have four different retailers for the new iPhone 14 cases.
- Everyone is ordering the casings from a production company in Atlanta.
- We assume that the weekly demand is almost constant and is equivalent to 780 sleeves.
- Please note the following table for order quantity, frequency and customer demand:

Seller	Number of orders	Frequency of orders	Weekly customer demand
W	570	every 2 weeks	285
X	120	every week	120
Y	525	every 3 weeks	175
Z	600	every 3 weeks	200
Σ			780



Bullwhip effect: example 2

Week	1	2	3	4	5	6	7	8	9	10	11	12
Retailer W	-	570	-	570	-	570	-	570	-	570	-	570
Retailer X	120	120	120	120	120	120	120	120	120	120	120	120
Retailer Y	-	525	-	-	525	-	-	525	-	-	525	-
Retailer Z	-	-	600	-	-	600	-	-	600	-	-	600
Demand Atlanta	120	1215	720	690	645	1290	120	1215	720	690	645	1290
Demand customer/week	780	780	780	780	780	780	780	780	780	780	780	780



Bullwhip effect: example 2

